# Spring AI: Beginner to Guru

Generative AI with Spring

## Overview of OpenAI

# OpenAI

- **OpenAI**
  - Private AI research company founded in 2015 by Elon Musk, Sam Altman, Greg Brockman and others
  - Original mission to develop safe and beneficial artificial general intelligence (AGI) for humanity
  - $1 billion initial endowment from founders/investors
  - Originally non-profit, shifted to "capped profit" model in 2019 (OpenAI LP)
  - Elon Musk left in 2018 due to potential conflict with Tesla
  - Leadership upheaval in late 2023 with firing/rehiring of Altman as CEO

# Detailed Evolution and Key Innovations - GPT

- **Generative Pre-trained Transformers (GPT) Series**
  - **GPT (2018):** The first in the series introduced the revolutionary transformer architecture to natural language processing (NLP), setting new benchmarks for text generation.
  - **GPT-2 (2019):** With significantly more parameters than its predecessor, GPT-2 demonstrated the ability to generate coherent and surprisingly human-like text passages, sparking discussions on AI's potential and ethics.

# Detailed Evolution and Key Innovations - GPT

- **Generative Pre-trained Transformers (GPT) Series**
  - **GPT-3 (2020):** A leap in scale and capability, GPT-3's 175 billion parameters enabled an unprecedented level of natural language understanding and generation, leading to wide-ranging applications from automated content creation to coding assistance.
  - **GPT-4 (2023):** Pushing the boundaries further, GPT-4 (estimated 1.76 trillion parameters) introduced multimodal capabilities, processing both text and image inputs, and showcased superior performance on complex tasks, including passing a simulated bar exam with scores in the top 10%.

# Detailed Evolution and Key Innovations - AI Art

- **Advances in AI-Generated Art and Imaging**
- **DALL·E (2021):** This model demonstrated the ability to generate diverse and creative images from textual descriptions, opening new avenues for AI in art and design.
- **DALL·E 2 and 3 (2022-2023):** Successive improvements on DALL·E, these versions offered more realistic and detailed image generation, enhancing the fidelity and versatility of visual content creation.

# Detailed Evolution and Key Innovations - Specialized

- **Specialized Models for Coding, Captioning, and Speech:**
  - **Codex:** Building on GPT-3, Codex was trained on billions of lines of code to assist in software development, powering tools like GitHub Copilot.
  - **Clip:** This neural network bridged the gap between visual and textual data, offering advanced capabilities in generating accurate captions for a wide array of images.
  - **Whisper (2022):** An advanced automatic speech recognition system, Whisper marked OpenAI's entry into processing and understanding spoken language across numerous languages and dialects.

# Revolutionizing Human-AI Interaction

- **Revolutionizing Human-AI Interaction**
  - **ChatGPT (2022):** This conversational model became a cultural phenomenon, offering nuanced and contextually aware responses across various domains, significantly impacting education, customer service, and more.
  - **Sora (2024):** A pioneering text-to-video model, Sora expanded the possibilities of content creation, allowing for the generation of detailed videos from textual prompts, showcasing OpenAI's continued innovation at the frontier of AI technology.

# Sora Example

- **Prompt:** Photorealistic closeup video of two pirate ships battling each other as they sail inside a cup of coffee.

# Limitations of GPT

- **Inaccuracies in Generated Content:** GPT models, including ChatGPT, have been known to produce "hallucinations" or factually incorrect information that appears coherent and plausible.
- **Outdated Information:** Due to the nature of their training data, GPT models may not provide the most current information, as they are limited to knowledge up to the point of their last training data update.
- **Lack of Contextual Depth:** While GPT models can generate human-like text, they may lack a deep understanding of complex or nuanced topics, leading to superficial responses.
- **Inability to Verify Facts:** GPT-generated content does not inherently distinguish between verified facts and fiction, potentially spreading misinformation.

# Criticisms of GPT

- **Ethical and Societal Concerns:** The shift from a non-profit to a "capped" for-profit model raised concerns about OpenAI's commitment to AI safety and openness, suggesting a potential focus on profit over public good.
- **Transparency Issues:** Critics have pointed out a lack of transparency in OpenAI's operations, particularly in relation to the detailed workings and training data of GPT models, diverging from initial commitments to open AI development.
- **Potential for Misuse:** The capabilities of GPT models, especially in generating coherent text, have raised concerns about their misuse in creating fake news, impersonating individuals, and other deceptive practices.

# Criticisms of GPT

- **Impact on Intellectual Property:** The training of GPT models on vast datasets scraped from the internet, including copyrighted materials, has led to lawsuits alleging copyright infringement, raising questions about the ethical use of data in training AI.
- **Data Privacy and Regulation:** There have been concerns and legal actions related to potential violations of data privacy regulations, particularly regarding the collection and use of personal data in training and deploying GPT models.

# Using ChatGPT

- Using ChatGPT is very easy
- **Disclaimer**: Content of this Presentation was generated by ChatGPT
  - Uploaded Wikipedia article on OpenAI to ChatGPT to provide current data
  - Used several prompts to generate bullet point summaries
  - Applied formatting and several minor edits to generated content

# SPRING FRAMEWORK GURU